AVR Microcontroller Laboratory – 10 Experiments in Embedded C

Peripherals, interrupts and buses on the ATmega16/32 (CodeVisionAVR / Atmel Studio + Proteus)

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Purpose

Ten laboratory experiments covering the peripherals of an 8-bit AVR microcontroller end to end: write the embedded C driver, configure the peripheral at register level, simulate the circuit in Proteus and validate the behaviour on the schematic. The experiments progress from GPIO up to interrupt-driven serial buses.

Experiments

- **1 – Digital I/O ports:** reading switches and driving LEDs through the port registers.
- **2 – Seven-segment displays:** multiplexed display driving with a modular source layout.
- **3 – Alphanumeric LCD and keypad:** LCD initialisation and printing, 4x4 keypad scanning, and interaction between the two.
- **4 – Timers and interrupts:** timer configuration, precise delays and interrupt service routines.
- **5 – Motor drives:** DC and stepper motor control (direction, speed).
- **6 – ADC:** the same conversion implemented both by polling and with interrupts, and the two approaches compared.
- **7 – UART:** initialisation, interrupt-driven transmission/reception and printing received text on the LCD.
- **8 – Graphic LCD:** 128x64 display with bitmap rendering, a clock and a dot-matrix font.
- **9 – SPI:** master/slave communication between two microcontrollers with LCD output (group exercise).
- **10 – I2C (TWI) and EEPROM:** master/slave I2C between an ATmega16 and an ATmega32, plus read/write of a 24LC32 EEPROM.

Verification and tools

Embedded C written in CodeVisionAVR and Atmel Studio, simulated in Proteus with the in-circuit AVR model; each experiment keeps its project files, its schematic (`.pdsprj`) and the compiled `.hex` so the simulation runs as submitted. My written reports for experiments 1, 3 and 5 are included, together with screen captures of the Proteus schematics and a compressed screen recording per experiment.

Repository contents

One folder per experiment (`exp01-io-port` … `exp10-i2c-eeprom`) with the C sources, project files, Proteus schematics, firmware images and demo videos; `docs/` with this overview.

Skills demonstrated

Embedded C on an 8-bit AVR; register-level peripheral programming (GPIO, timers, ADC, UART, SPI, I2C/TWI); interrupt-driven design and ISRs; display drivers for 7-segment, alphanumeric and graphic LCDs; motor control; master/slave multi-microcontroller buses; EEPROM interfacing; the CodeVisionAVR and Atmel Studio toolchains; simulation-based validation in Proteus.